Einstein's General Theory of Relativity

Lecture 4: Tensor Algebra, Metrics, and Spacetime

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Chapter 1

Tensor Algebra, Metrics, and Spacetime

1.1 The Minimum Is Technical

The theoretical minimum is not a list of conclusions stated in words. It is the least mathematics needed to move reliably to the next subject, which in this case is general relativity and then cosmology. No advanced machinery is required tonight, but the calculations must be followed patiently. Index placement, transformation laws, and contractions are not decoration; they are the grammar in which curved geometry is written.

Let x^m and y^m be two coordinate systems on the same space. A covariant vector transforms as

$$V_m^{(y)}(y) = \frac{\partial x^n}{\partial y^m} V_n^{(x)}(x), \tag{1.1}$$

whereas a contravariant vector transforms as

$$W_{(y)}^m(y) = \frac{\partial y^m}{\partial x^n} W_{(x)}^n(x). \tag{1.2}$$

The free index m appears on both sides and belongs to the y description. The repeated n is summed and belongs to the x description.

A collection of numbers is not a vector merely because it can be arranged as an arrow. Temperature, pressure, and humidity are three scalars. Putting their numerical values along three axes does not make a geometric vector, because those values do not mix under a coordinate change according to Eq. (1.1) or Eq. (1.2). Transformation behavior defines the object.

The two laws come directly from calculus. A scalar gradient satisfies

$$\frac{\partial \phi}{\partial y^m} = \frac{\partial x^n}{\partial y^m} \frac{\partial \phi}{\partial x^n}, \tag{1.3}$$

so gradients are covariant. Coordinate differentials satisfy

$$dy^m = \frac{\partial y^m}{\partial x^n}\, dx^n, \tag{1.4}$$

so infinitesimal displacements are contravariant. These chain rules are the archetypes from which the terminology is abstracted.

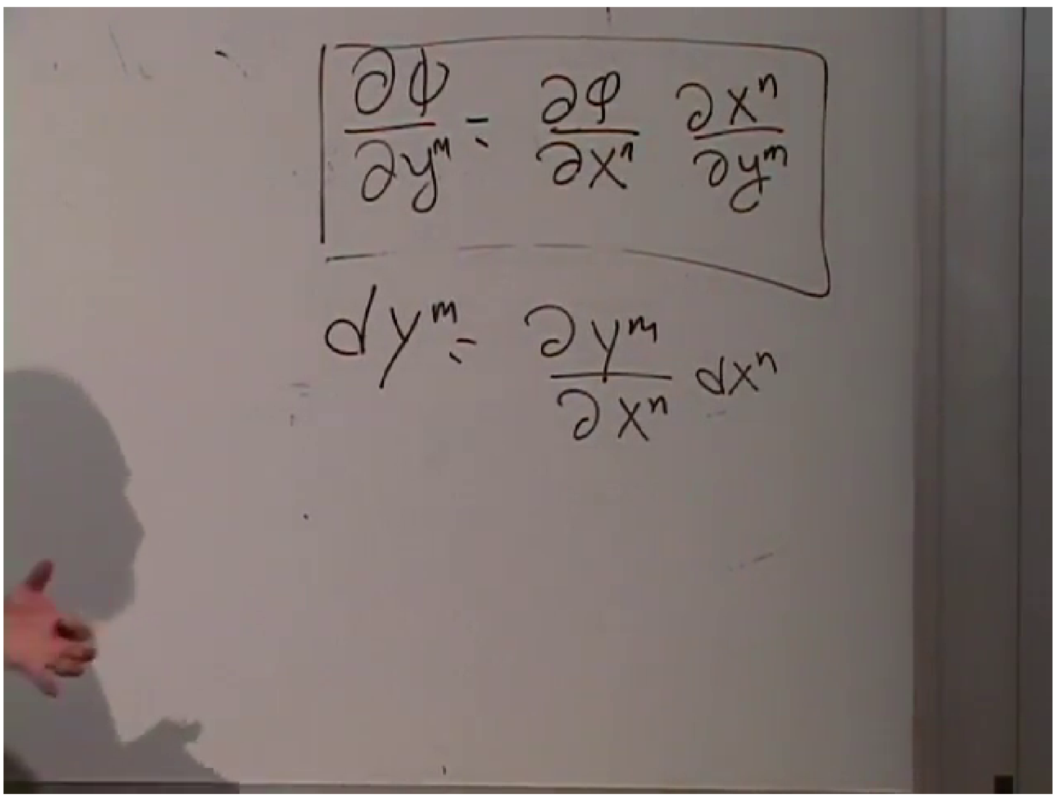

Figure 1.1: The two chain-rule archetypes: a scalar gradient transforms covariantly, while a differential displacement transforms contravariantly.

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Question from the audience. If an object satisfies one vector transformation law, must it also satisfy the other? [a]

Response. No. Before a metric is introduced, a covector such as $\partial_m \phi$ and a vector such as dx^m are different kinds of object. The same letter should not be used to identify them. The metric will later provide a one-to-one map between their component descriptions.

[a] Lecture timestamp: 00:07:24–00:08:33.

Coordinates conventionally carry upper indices. A coordinate in the denominator of a derivative therefore contributes a lower index, while a coordinate differential naturally carries an upper one. The convention is arbitrary, but once adopted it makes the Jacobian patterns difficult to misread.

1.2 Higher-Rank Tensors and Contraction

Each tensor index transforms independently. For a tensor with two lower indices,

$$T^{(y)}_{mr} = \frac{\partial x^n}{\partial y^m}\frac{\partial x^s}{\partial y^r} T^{(x)}_{ns}. \tag{1.5}$$

For one upper and one lower index,

$$\left(T^m{}_r\right)_{(y)} = \frac{\partial y^m}{\partial x^n}\frac{\partial x^s}{\partial y^r}\left(T^n{}_s\right)_{(x)}. \tag{1.6}$$

The general rule is one direct Jacobian for every upper index and one inverse Jacobian for every lower index.

Question from the audience. May upper and lower indices be mixed on the same tensor? [a]

Response. Yes. A mixed tensor is perfectly legitimate. Each index retains its own transformation rule, as Eq. (1.6) shows.

[a] Lecture timestamp: 00:15:15–00:16:18.

The next operation is *contraction.* Identify one upper index with one lower index and sum:

$$T^m{}_m = T^1{}_1 + T^2{}_2 + \cdots . \tag{1.7}$$

No free index remains, so the result should be a scalar. The transformation law verifies it:

$$\begin{aligned}\left(T^m{}_m\right)_{(y)} &= \frac{\partial y^m}{\partial x^r}\frac{\partial x^s}{\partial y^m}\left(T^r{}_s\right)_{(x)} \\ &= \delta^s{}_r\, T^r{}_s{}^{(x)} = \left(T^r{}_r\right)_{(x)}. \end{aligned} \tag{1.8}$$

The paired Jacobians are inverse matrices. Their product is the identity, represented by the Kronecker delta.

The same argument works at any rank. Contracting one upper with one lower index removes those two indices and leaves a tensor whose remaining indices transform normally. A familiar special case is the inner product

$$V^m W_m, \tag{1.9}$$

which is a scalar.

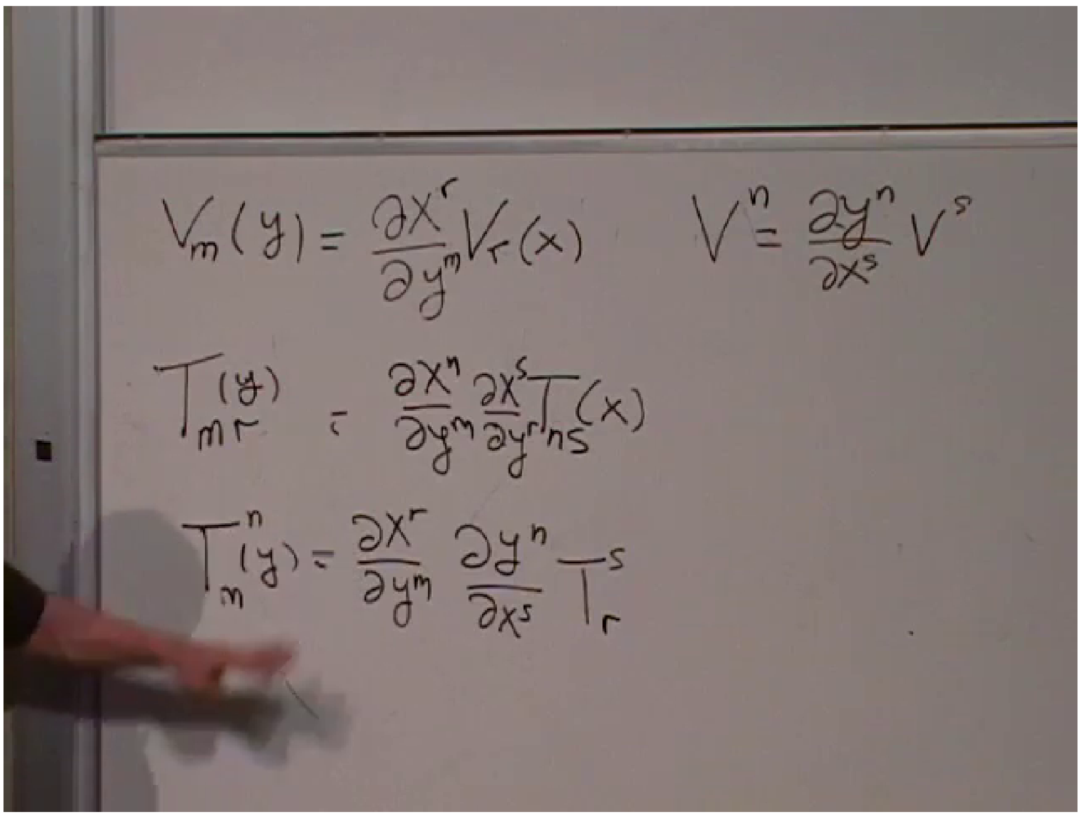

Figure 1.2: Covariant and mixed rank-two transformation laws. The completed board displays one Jacobian factor for each tensor index.

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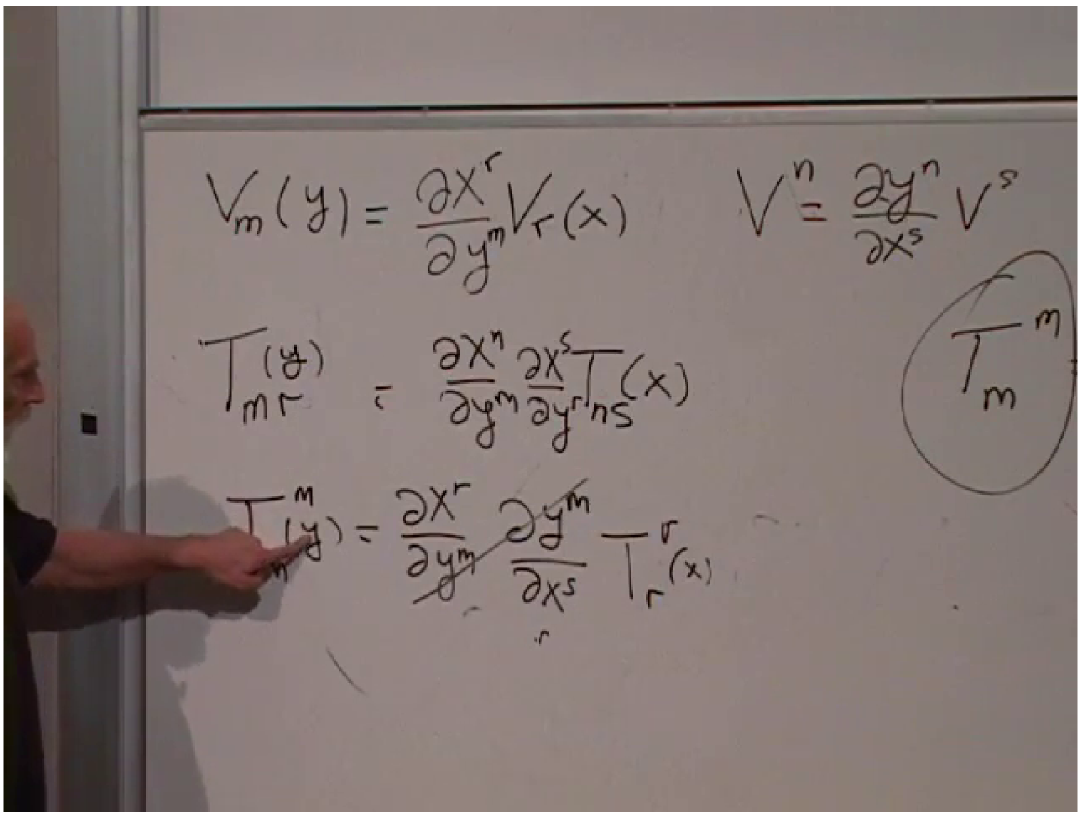

Figure 1.3: Contraction of a mixed tensor. The circled trace has no free indices and is unchanged by the coordinate transformation.

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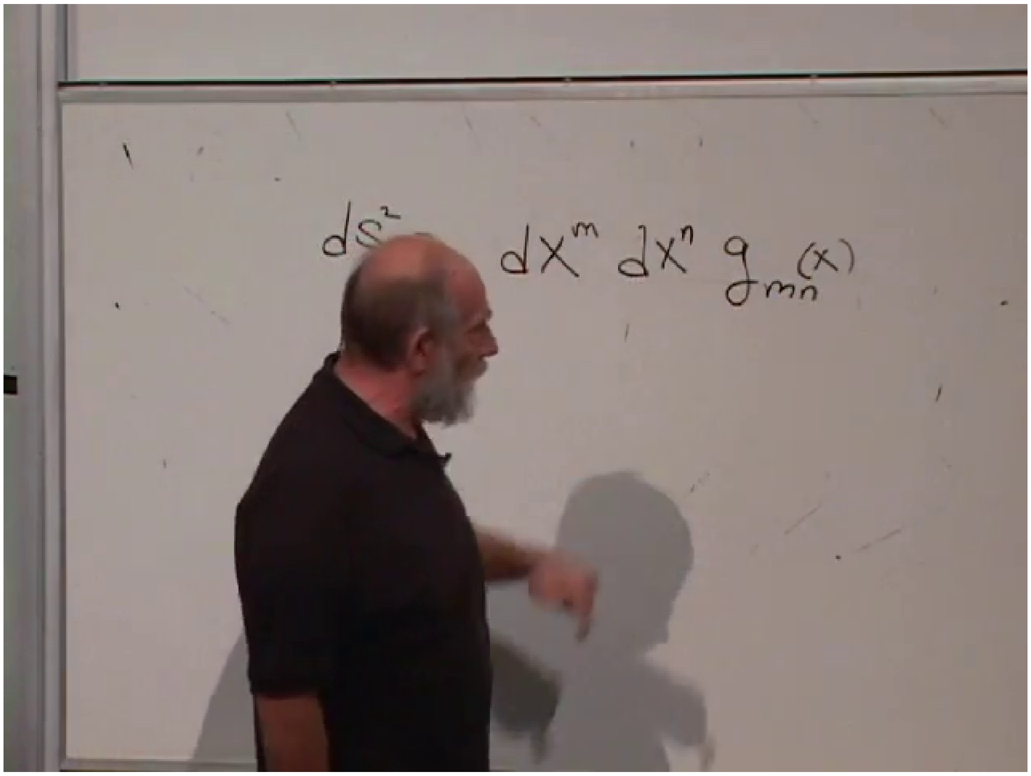

Figure 1.4: The metric tensor enters as the bilinear coefficient that turns an infinitesimal displacement into its squared length.

Lecture frame: 00:28:35.

Question from the audience. Does contraction imply that the original tensor was a scalar times the lower-rank tensor that remains? [a]

Response. No. Contraction constructs a new object and generally discards information. A special tensor may factor, but no such factorization follows from contraction alone.

[a]Lecture timestamp: 00:25:00–00:26:50.

Question from the audience. Does a two-index tensor necessarily represent a plane? [a]

Response. Sometimes a rank-two object admits that geometric interpretation, but rank by itself does not force it. The metric tensor is the important example here: its two indices encode the bilinear rule for distance.

[a]Lecture timestamp: 00:26:50–00:27:35.

1.3 The Metric from the Scalar Line Element

For two neighboring points separated by dx^m, define the squared distance

$$\boxed{ds^2 = g_{mn}(x)\, dx^m dx^n.} \tag{1.10}$$

Both vector indices are contracted against the two lower metric indices, so ds^2 is a scalar.

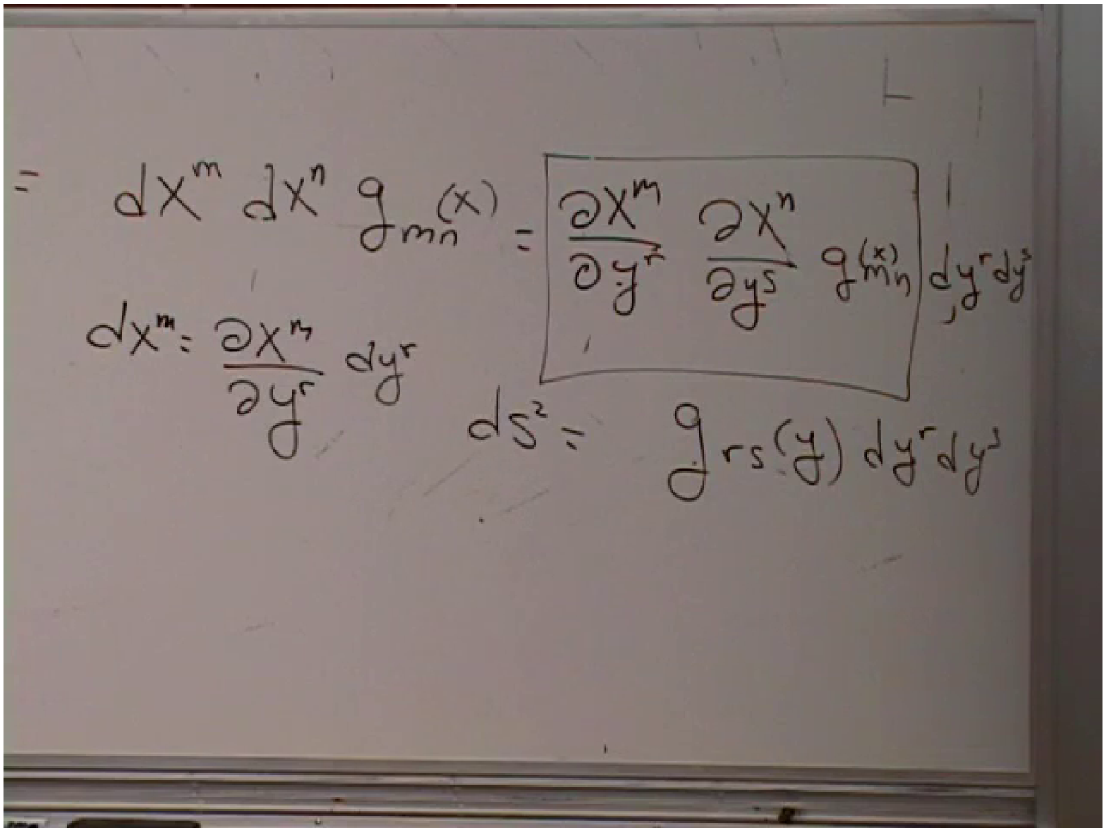

Figure 1.5: The line element rewritten in y-coordinates. The boxed coefficient of $dy^r dy^s$ is the transformed metric $g^{(y)}_{rs}$.

Lecture frame: 00:31:25.

To prove that g_{mn} is a tensor, rewrite the same scalar in the y-coordinates:

$$dx^m = \frac{\partial x^m}{\partial y^r}\, dy^r. \tag{1.11}$$

Substitution gives

$$\begin{aligned} ds^2 &= g^{(x)}_{mn} \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s} dy^r dy^s \\ &= g^{(y)}_{rs}\, dy^r dy^s, \end{aligned} \tag{1.12}$$

and therefore

$$\boxed{g^{(y)}_{rs} = \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s} g^{(x)}_{mn}.} \tag{1.13}$$

This is exactly the covariant rank-two law.

Question from the audience. Is ds^2 really the same in every coordinate system? [a]

Response. Yes. It is one scalar interval written in two coordinate languages. Its invariance is not a result discovered after the transformation law; it is the premise that forces Eq. (1.13).

[a] Lecture timestamp: 00:45:25–00:46:25.

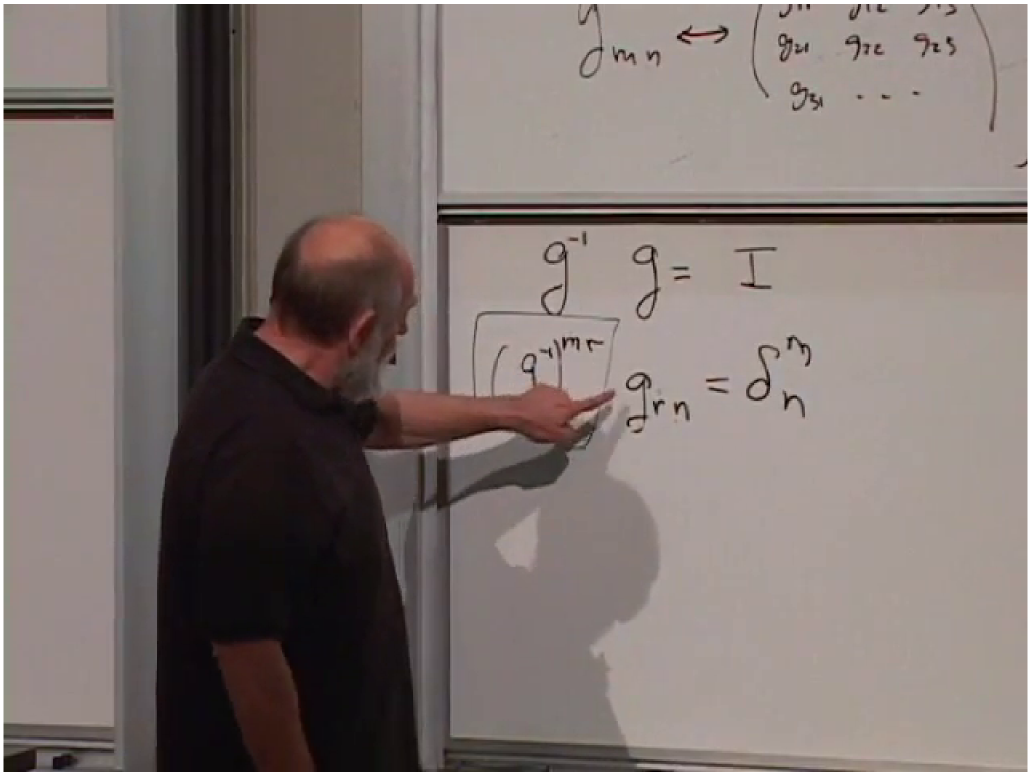

Figure 1.6: Matrix inversion translated into tensor notation. The identity tensor has one upper and one lower index.

Lecture frame: 00:36:55.

1.4 The Inverse Metric

At a fixed point, the components g_{mn} form a matrix. The metric used for geometry is symmetric and nondegenerate:

$$g_{mn} = g_{nm}, \qquad \det(g_{mn}) \neq 0. \tag{1.14}$$

It therefore has an inverse matrix. Denote the inverse components by g^{mn}. Matrix multiplication becomes

$$\boxed{g^{mr} g_{rn} = \delta^m{}_n.} \tag{1.15}$$

The inverse components transform as a contravariant rank-two tensor. Thus g_{mn} and g^{mn} contain the same geometric information.

Question from the audience. Should both indices on the Kronecker delta be lower? [a]

Response. Not in the invariant identity-map notation used here. $\delta^m{}_n$ carries one upper and one lower index because it maps a vector space to itself. In an orthonormal Cartesian calculation one often writes δ_{mn}, but that restricted convenience should not be confused with the general identity tensor.

[a] Lecture timestamp: 00:33:20–00:34:32.

In polar coordinates on the plane,

$$ds^2 = dr^2 + r^2 d\theta^2, \tag{1.16}$$

so

$$(g_{mn}) = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \qquad (g^{mn}) = \begin{pmatrix} 1 & 0 \\ 0 & r^{-2} \end{pmatrix}. \tag{1.17}$$

For a diagonal matrix, inversion simply reciprocates each diagonal entry.

Question from the audience. Is the angular metric component r or r^2? [a]

Response. It is r^2. The coordinate differential is already $d\theta$; the squared physical arc length is $(r\,d\theta)^2 = r^2 d\theta^2$.

[a] Lecture timestamp: 00:37:30–00:38:24.

Question from the audience. Can a tensor fail to have an inverse? [a]

Response. A general rank-two tensor can be singular. A metric, by definition, is nondegenerate and therefore has no zero eigenvalue. Symmetry alone is not enough; nondegeneracy is the condition that guarantees the inverse. The same statement holds for a Lorentzian metric despite its minus signs.

[a] Lecture timestamp: 00:44:00–00:45:25.

Only the symmetric part contributes to the quadratic form. Indeed,

$$g_{12}\,dx^1 dx^2 + g_{21}\,dx^2 dx^1 = (g_{12} + g_{21})\,dx^1 dx^2. \tag{1.18}$$

An antisymmetric part would cancel because the coordinate differentials commute.

Question from the audience. Must g_{12} and g_{21} be equal? [a]

Response. For the metric, we take them equal without loss of geometric information. Any antisymmetric difference contributes nothing to ds^2, so retaining it would add redundant components rather than new distance data.

[a] Lecture timestamp: 00:46:30–00:49:00.

1.5 Raising and Lowering Indices

The metric now links the two vector descriptions:

$$V_n = g_{nm} V^m, \qquad V^m = g^{mn} V_n. \tag{1.19}$$

The first equation lowers an index and the second raises it. Because the metric is invertible, either set of components determines the other. In geometric terms, V_n is the covector obtained by taking the metric inner product of V with each coordinate basis vector. In an orthonormal basis this reduces to the familiar projection picture.

Indices on higher-rank tensors are handled one at a time. For example,

$$T^m{}_n = g^{mr} T_{rn}, \tag{1.20}$$

raises the first index of a covariant rank-two tensor.

Apply the same operation to the displacement:

$$dx_m = g_{mn}\,dx^n. \tag{1.21}$$

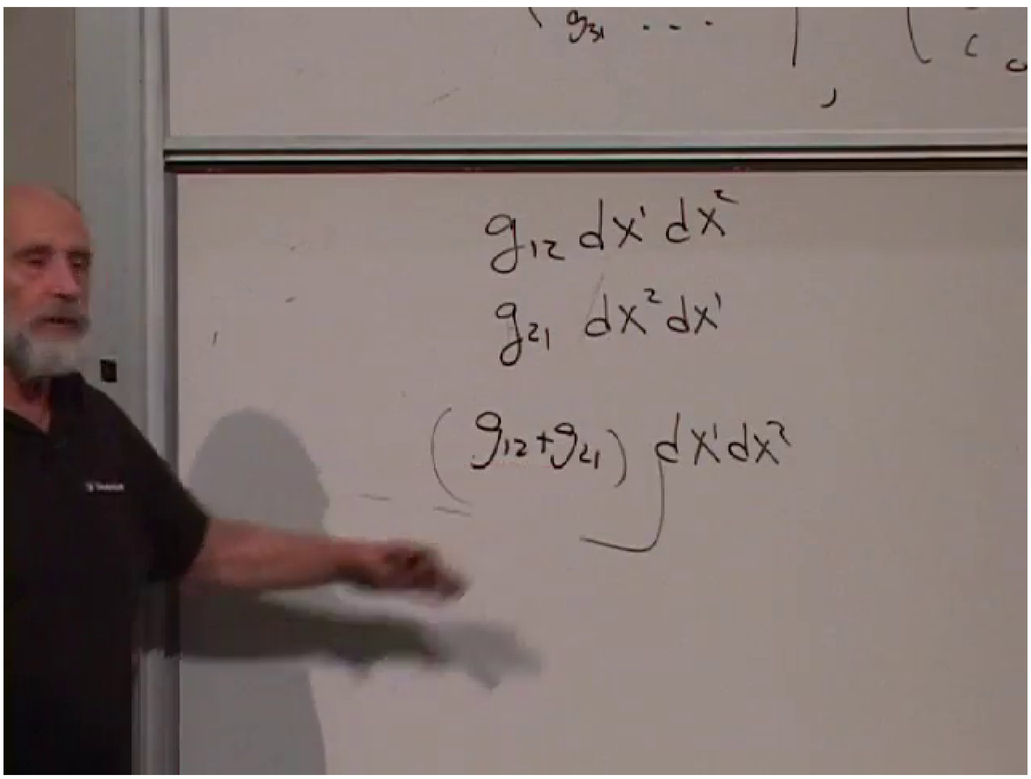

Figure 1.7: The two off-diagonal contributions combine through $g_{12} + g_{21}$, showing why the line element depends only on the symmetric part of the metric.

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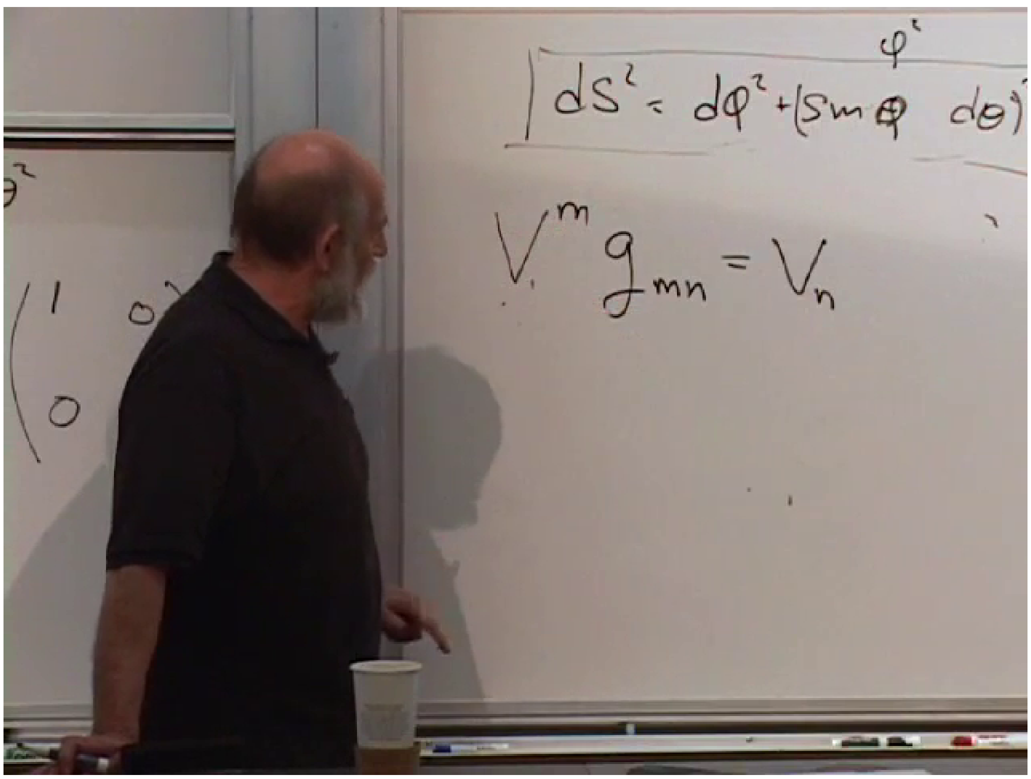

Figure 1.8: Lowering a vector index with g_{mn}. The metric turns the contravariant components V^m into covariant components V_n.

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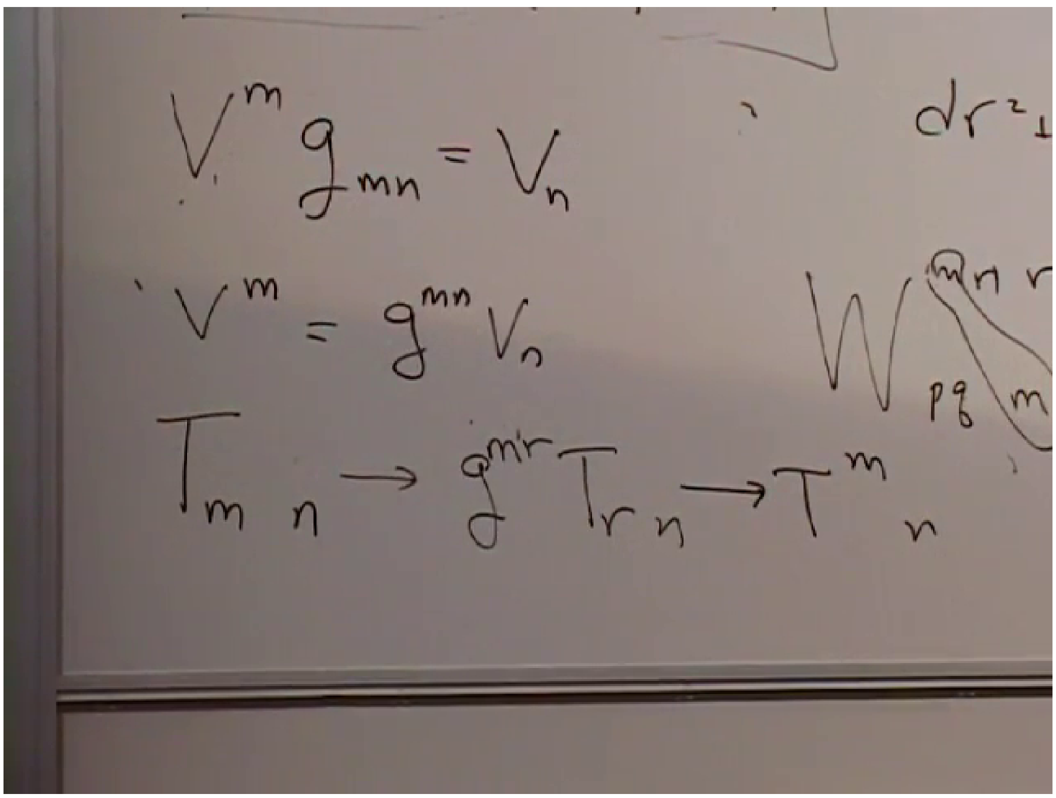

Figure 1.9: The vector inverse relation and the corresponding operation of raising one index on a rank-two tensor.

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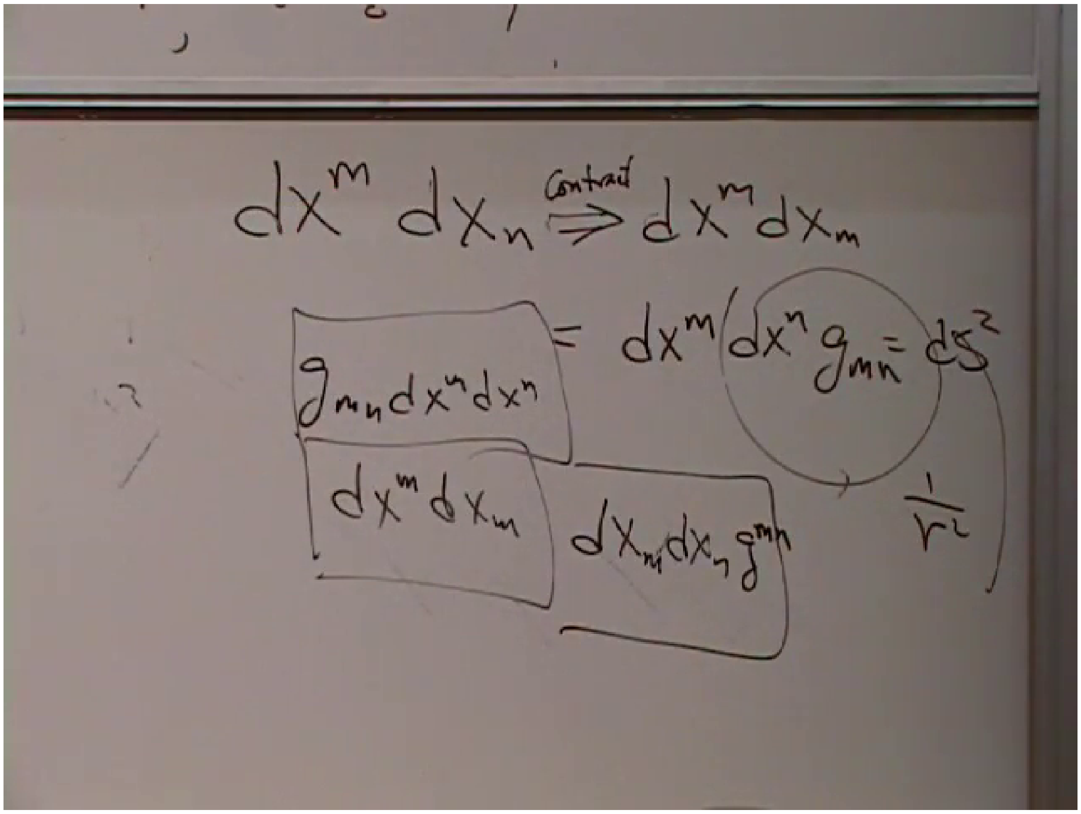

Figure 1.10: The completed board comparison of the three contracted forms of the same scalar interval.

Lecture frame: 00:58:15.

The squared interval then has three equivalent forms,

$$\boxed{ds^2 = g_{mn}\, dx^m dx^n = dx^m dx_m = g^{mn}\, dx_m dx_n.} \tag{1.22}$$

Question from the audience. Was the first expression on the board meant as an equality between $dx^m dx_n$ and its contraction? [a]

Response. No. That equals sign is withdrawn in the discussion. One first identifies an upper and lower index and sums; only after carrying out that contraction does one obtain the scalar $dx^m dx_m = ds^2$.

[a]Lecture timestamp: 00:56:30–00:57:45.

Question from the audience. What is the relation between the upper- and lower-index metrics in these formulas? [a]

Response. They are inverse matrices, related by Eq. (1.15). For a diagonal metric the entries are reciprocals. With off-diagonal components one performs an ordinary matrix inverse.

[a]Lecture timestamp: 00:58:20–00:59:55.

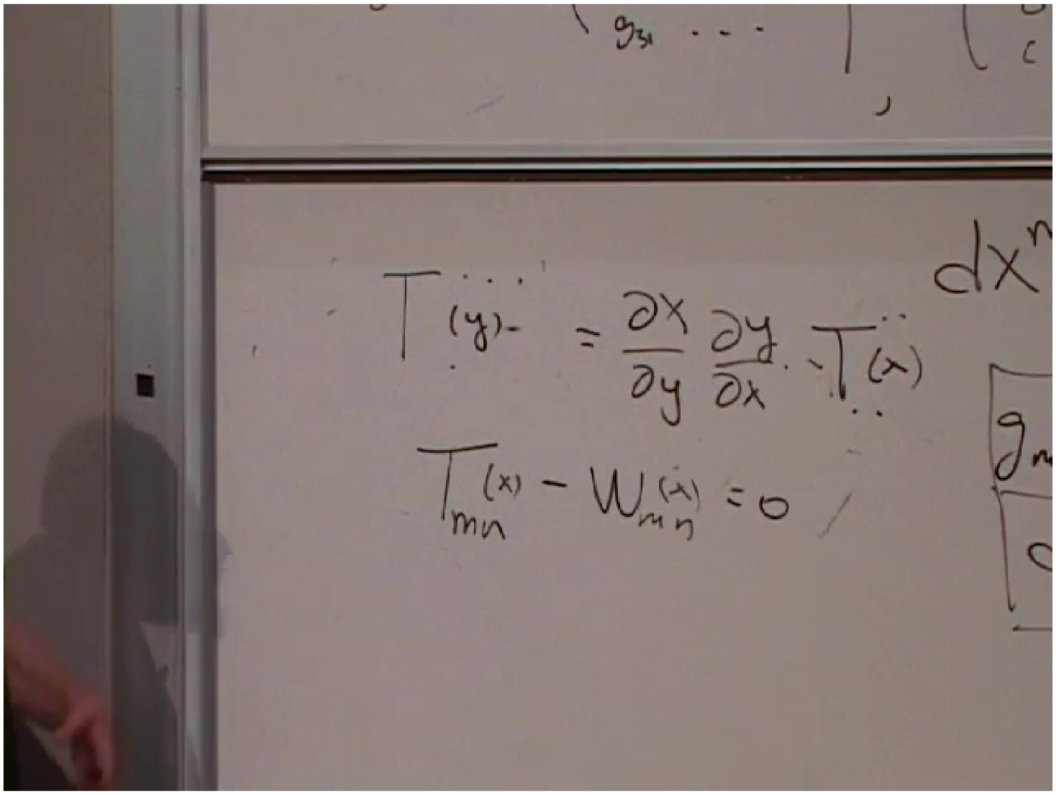

Figure 1.11: A tensor equality recast as the vanishing of the difference. Once the difference is the zero tensor in one chart, it is zero in every chart.

Lecture frame: 01:02:40.

1.6 Tensor Equations and the Deferred Calculus

Tensors can be added only when their index types agree. The sum $T_{mn} + W_{mn}$ transforms coherently; a formal sum of T_{mn} and W^{mn} does not.

If every component of a tensor vanishes in one coordinate system, every component vanishes in all coordinate systems, because each transformed component is a linear combination of zeros. Hence if

$$T_{mn} = W_{mn} \tag{1.23}$$

in one chart, then

$$T_{mn} - W_{mn} = 0 \tag{1.24}$$

there and therefore everywhere. Tensor equality is geometric even though the separate components depend on coordinates.

This completes tensor algebra, not tensor calculus. Ordinary partial derivatives of tensor components do not generally transform as tensor components, because differentiating the Jacobians produces extra terms. The corrected derivative will be introduced later. For now the lecture deliberately turns from space to spacetime.

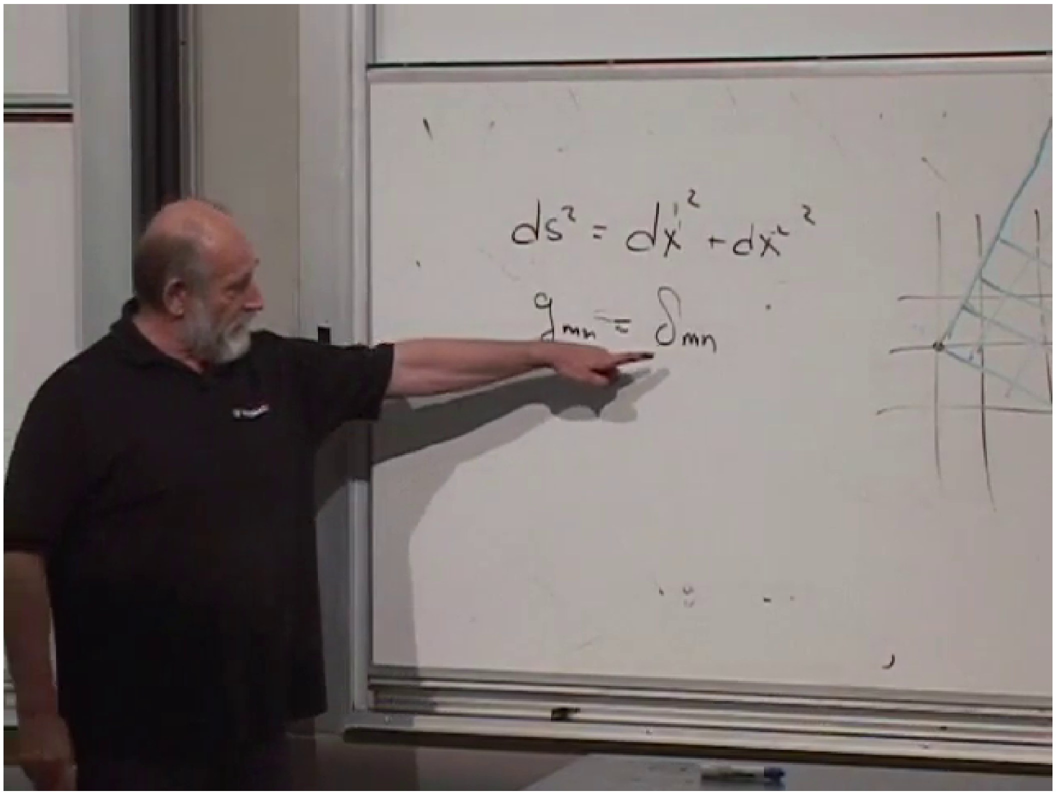

Figure 1.12: Two Cartesian frames on the same flat plane together with the Euclidean interval they preserve.

Lecture frame: 01:09:25.

1.7 From Cartesian Space to Proper Time

Even flat Euclidean space has many Cartesian frames. They may be translated or rotated relative to one another. In two dimensions,

$$ds^2 = (dx^1)^2 + (dx^2)^2 = \delta_{mn}\, dx^m dx^n. \tag{1.25}$$

Cartesian-to-Cartesian transformations preserve this Pythagorean form. In that restricted setting the identity metric makes upper and lower components numerically equal.

Spacetime has one time coordinate and three spatial coordinates. In one space dimension, a Euclidean-looking sum $dt^2 + dx^2$ is not preserved by Lorentz transformations. The invariant quantity is the proper-time interval

$$\boxed{d\tau^2 = dt^2 - dx^2} \qquad (c = 1). \tag{1.26}$$

Setting $c = 1$ makes time and distance use the same unit. Keeping different units would be like measuring one Cartesian direction in feet and another in inches: conversion factors would obscure the geometric symmetry. If c is restored,

$$d\tau^2 = dt^2 - \frac{dx^2}{c^2}, \tag{1.27}$$

when $d\tau$ is measured in time units.

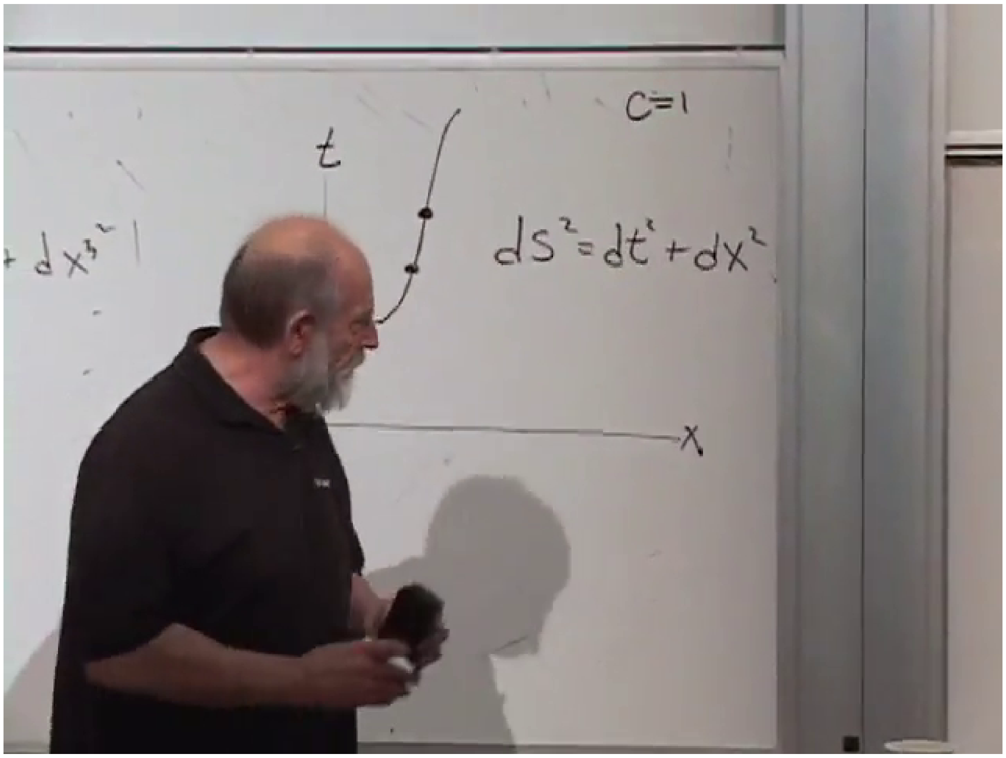

Figure 1.13: Two nearby events on a worldline and the invariant proper-time interval $d\tau^2 = dt^2 - dx^2$.
Lecture frame: 01:13:35.

Question from the audience. How can particle physicists express length, time, and mass in electron volts? [a]

Response. Setting $c = 1$ identifies time units with length units. Setting $\hbar = 1$ additionally relates inverse length and inverse time to energy or mass. Only $c = 1$ is needed in the present relativistic discussion; the $\hbar = 1$ convention belongs to quantum theory.

[a] Lecture timestamp: 01:16:00–01:17:32.

In three spatial dimensions,

$$\boxed{d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2.} \tag{1.28}$$

Special relativity may be characterized as the theory of inertial coordinate transformations that preserve this indefinite interval.

1.8 General Spacetime Coordinates

Accelerated coordinates have already shown that curvilinear coordinates are useful in spacetime even before genuine gravitational curvature is introduced. Write

$$x^\mu = (x^0, x^1, x^2, x^3), \qquad x^0 = t. \tag{1.29}$$

Greek indices run over spacetime coordinates. In arbitrary coordinates the proper-time element takes the tensorial form

$$\boxed{d\tau^2 = g_{\mu\nu}(x)\, dx^\mu dx^\nu.} \tag{1.30}$$

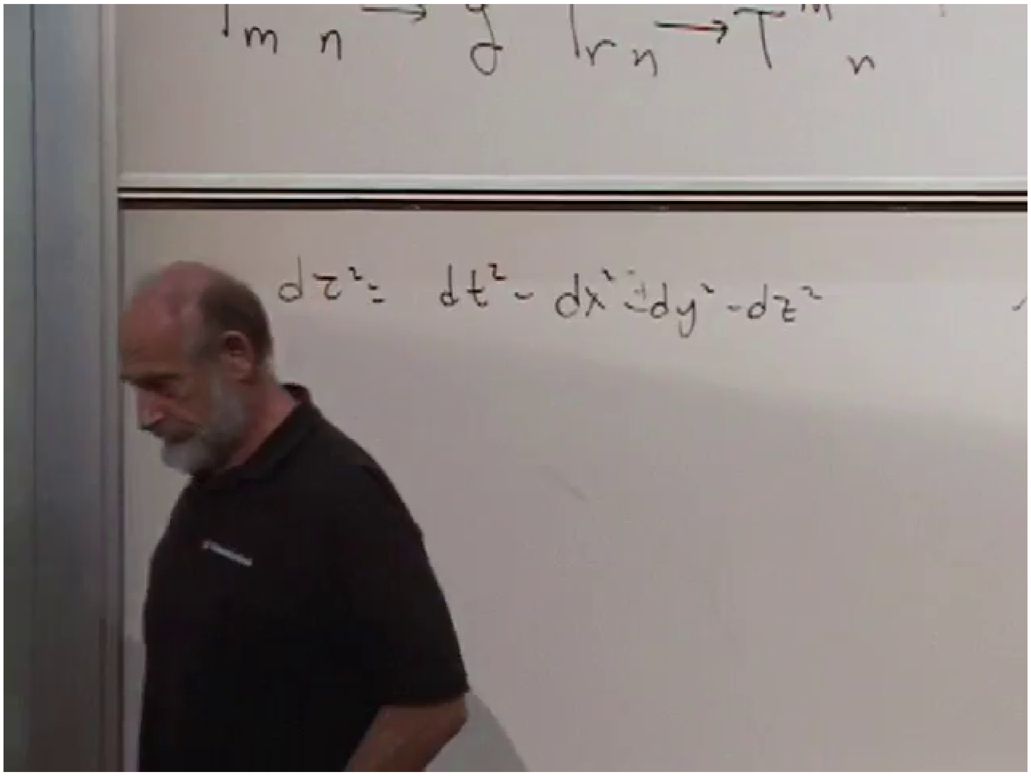

Figure 1.14: The proper-time interval extended to one time and three spatial dimensions.
Lecture frame: 01:18:40.

Question from the audience. Why number time by 0, and may a spacetime have more than one time coordinate? [a]

Response. Starting with $x^0 = t$ is a convention that leaves x^1, x^2, x^3 for the spatial coordinates. Higher-dimensional models may add spatial coordinates, but the Lorentzian geometry used here has one timelike direction. Multiple-time theories are outside the present framework.

[a] Lecture timestamp: 01:20:00–01:21:50.

In Minkowski coordinates,

$$\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1), \tag{1.31}$$

and

$$d\tau^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu. \tag{1.32}$$

The symbol $\eta_{\mu\nu}$ is the spacetime counterpart of the Cartesian δ_{mn}, but its signs encode a Lorentzian rather than Euclidean geometry.

Under every nonsingular real coordinate transformation, the numbers of positive and negative metric eigenvalues are preserved. With the convention used here, a physical four-dimensional spacetime metric has one positive and three negative directions.

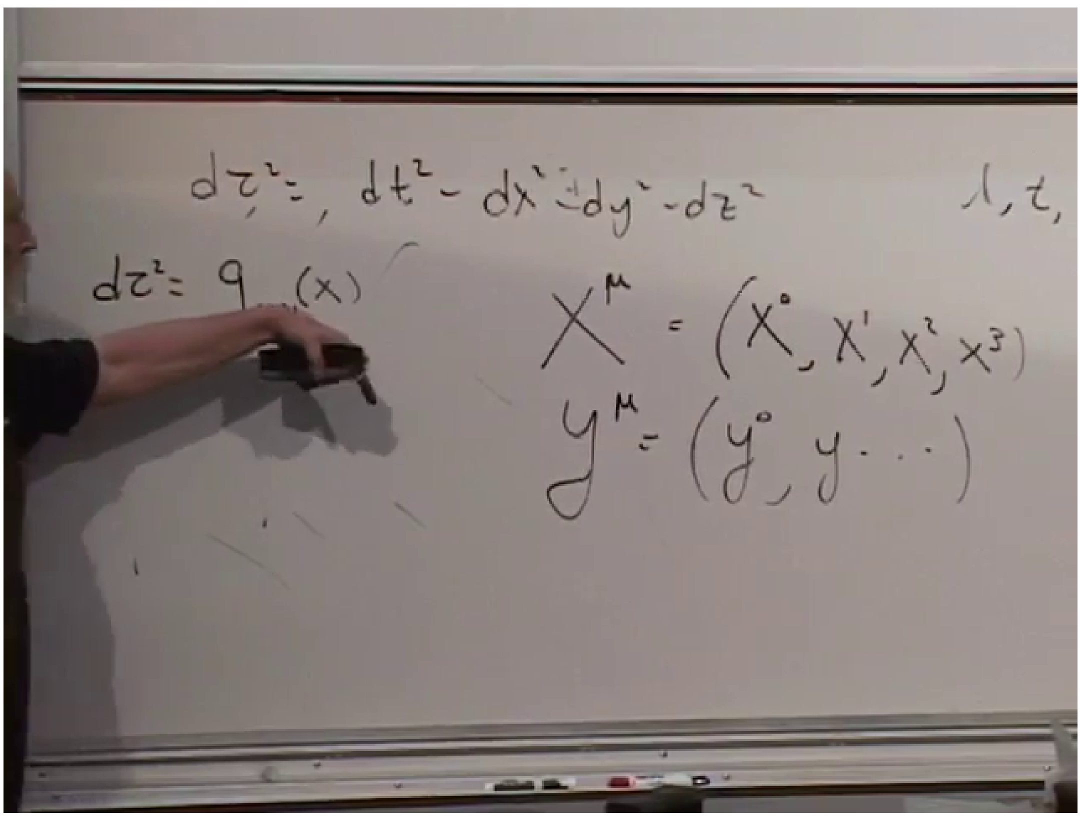

Figure 1.15: The general spacetime line element and the x^μ and y^μ coordinate labels.

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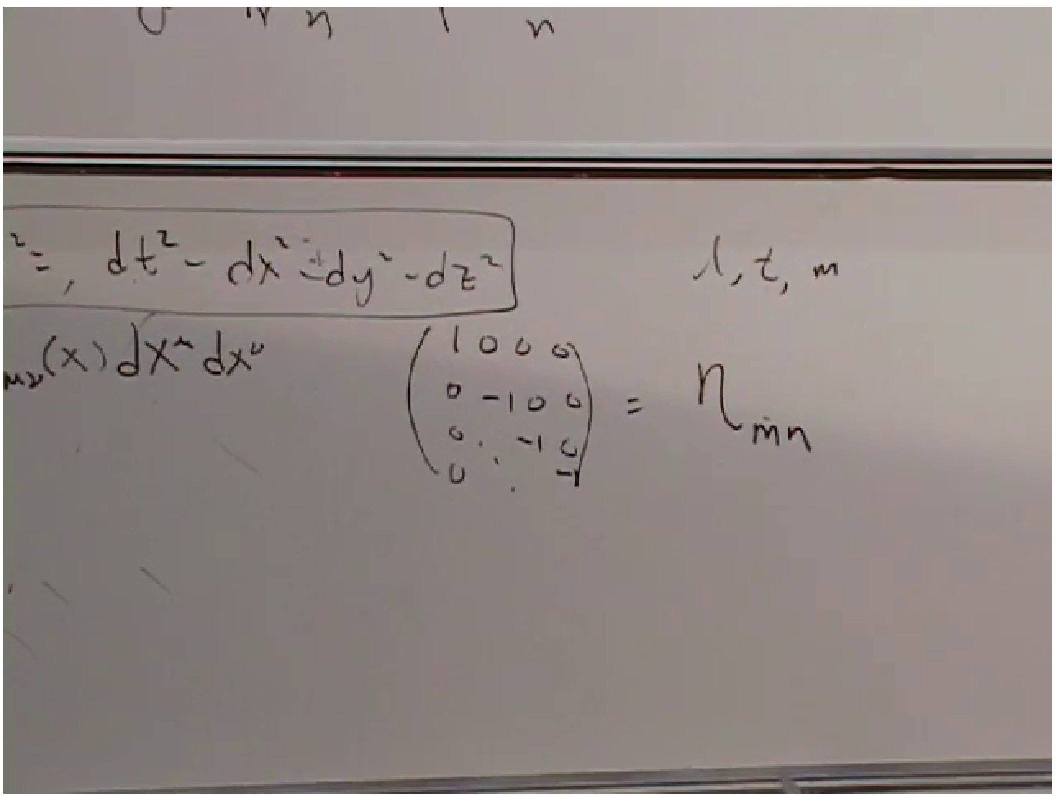

Figure 1.16: The Minkowski metric matrix with one positive and three negative diagonal entries.

Lecture frame: 01:26:10.

Question from the audience. Could a coordinate change simply interchange the labels x and t? [a]

Response. Labels may be exchanged, and arbitrary smooth coordinates may mix space and time. The metric components then change accordingly. What cannot be changed by a regular coordinate transformation is the Lorentzian signature: there remains one timelike and three spacelike directions.

[a]Lecture timestamp: 01:27:50–01:29:18.

Question from the audience. Is the Minkowski metric its own inverse? [a]

Response. Yes. Each diagonal entry is $+1$ or -1, and its reciprocal is itself, so $\eta^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$.

[a]Lecture timestamp: 01:29:18–01:30:00.

1.9 A First Expanding-Spacetime Metric

For a homogeneous and isotropic flat spatial geometry, choose comoving coordinates and write

$$(g_{\mu\nu}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -a^2(t) & 0 & 0 \\ 0 & 0 & -a^2(t) & 0 \\ 0 & 0 & 0 & -a^2(t) \end{pmatrix}. \tag{1.33}$$

The corresponding interval is

$$\boxed{d\tau^2 = dt^2 - a^2(t)\left(dx^2 + dy^2 + dz^2\right).} \tag{1.34}$$

The single function $a(t)$ is the scale factor. At a fixed cosmic time, two comoving points separated by coordinate distance $\Delta\ell_{\mathrm{com}}$ have physical distance

$$\Delta\ell_{\mathrm{phys}}(t) = a(t)\,\Delta\ell_{\mathrm{com}}. \tag{1.35}$$

If $a(t)$ grows, the physical separation grows even while the comoving coordinates remain fixed.

Question from the audience. What does it mean geometrically to say that the universe is expanding? [a]

Response. It means that the spatial part of the spacetime metric changes with time. In comoving coordinates, fixed coordinate separations correspond to physical distances proportional to $a(t)$. Expansion is therefore a statement about the metric, not motion through a preexisting external space.

[a]Lecture timestamp: 01:30:00–01:33:00.

Two-dimensional surfaces make the role of a changing metric coefficient visible. A rotationally symmetric surface can be written

$$ds^2 = dr^2 + f^2(r)\,d\theta^2. \tag{1.36}$$

The circumference of the circle at fixed r is $2\pi f(r)$.

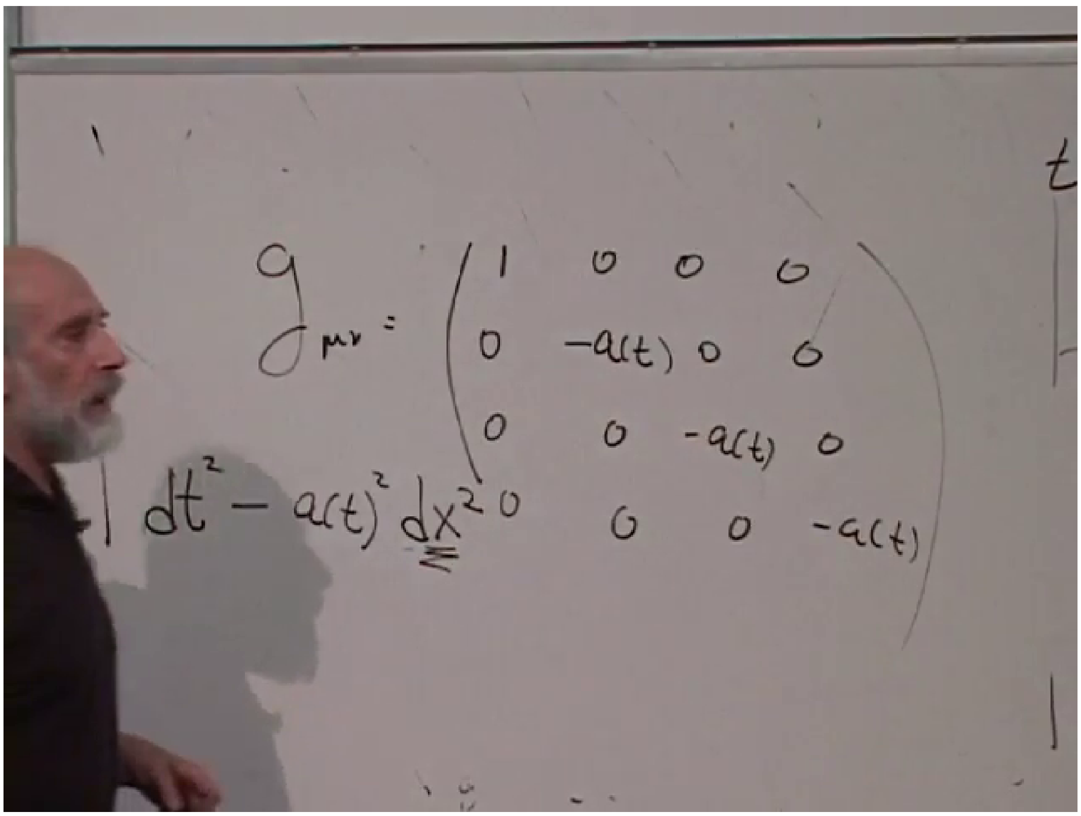

Figure 1.17: A homogeneous time-dependent metric with one common spatial scale factor controlling all three spatial directions.

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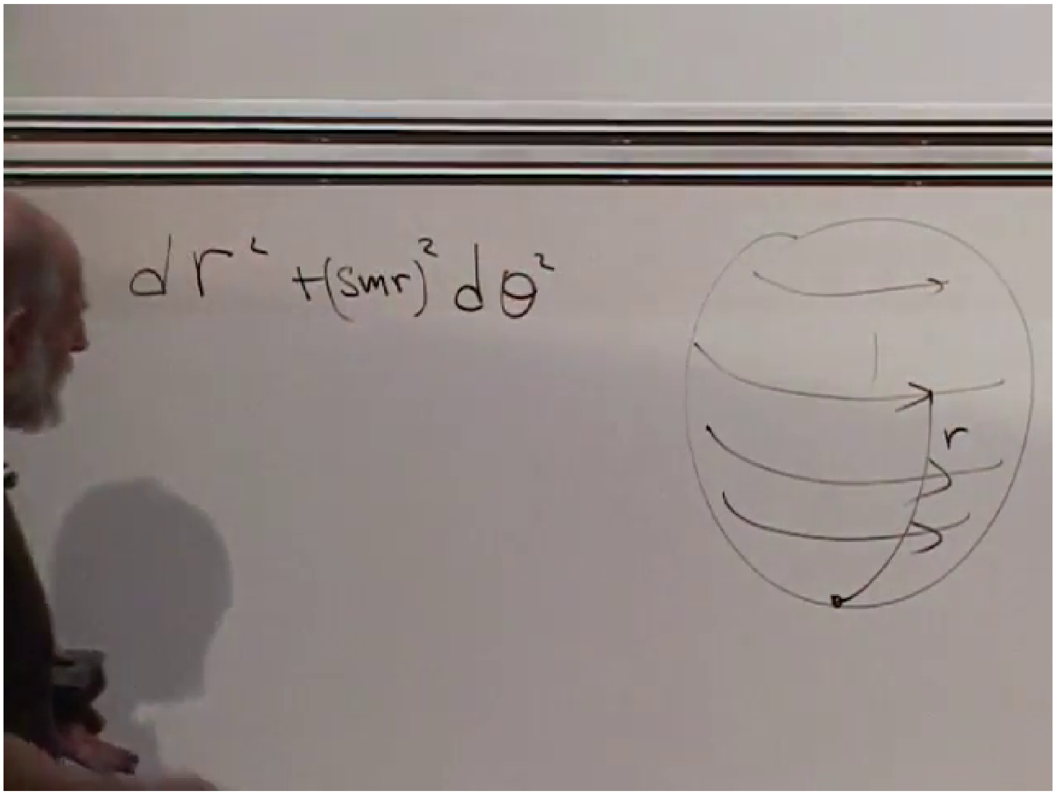

Figure 1.18: The sphere and its metric. The factor $\sin r$ controls the circumference of circles at fixed polar distance r.

Lecture frame: 01:35:30.

For the flat plane, $f(r) = r$. For a unit sphere,

$$f(r) = \sin r, \qquad ds^2 = dr^2 + \sin^2 r\, d\theta^2. \tag{1.37}$$

The circles grow from one pole to the equator and then shrink toward the other pole.

For $f(r) = e^r$,

$$ds^2 = dr^2 + e^{2r} d\theta^2, \tag{1.38}$$

the circumference grows exponentially and the surface has a horn-like profile. More complicated growing and oscillating choices of $f(r)$ produce correspondingly complicated surfaces.

These surfaces vary with position r; the cosmological metric varies with time t. The analogy is not yet a curvature calculation. Its purpose is to make metric coefficients readable as geometry. The next steps are parallel transport, curvature, and eventually Einstein's field equations.

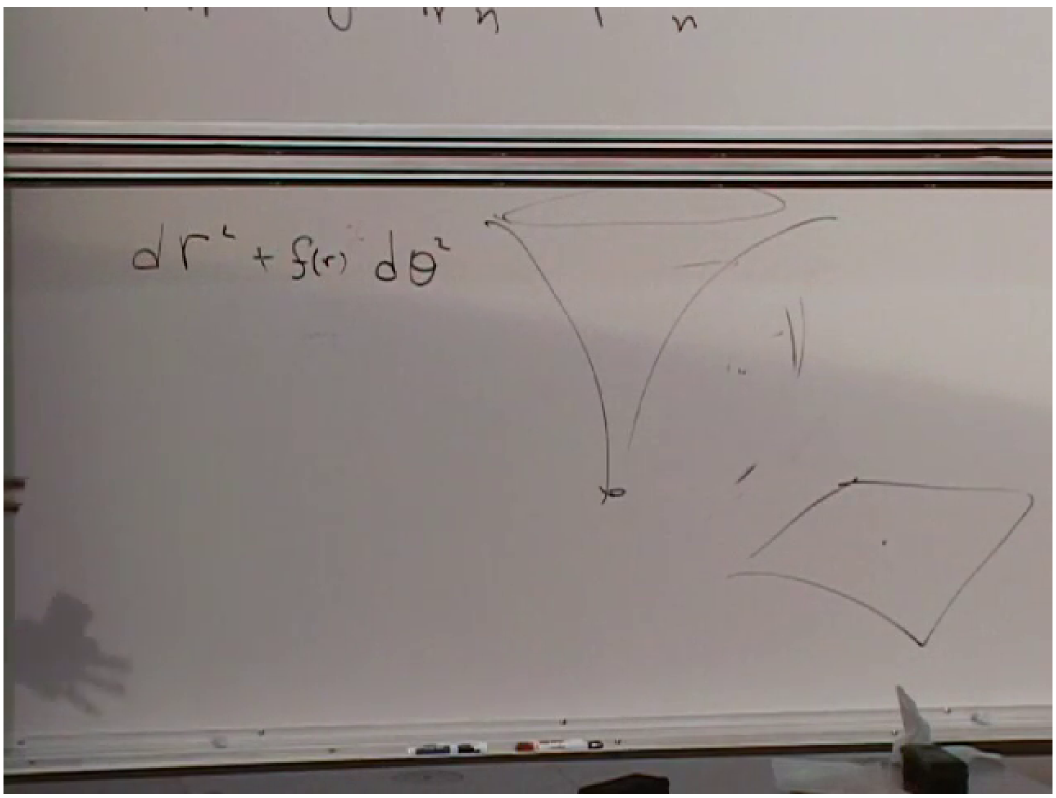

Figure 1.19: A general radial profile and a horn-like surface. The coefficient of $d\theta^2$ determines how circumference changes along r.

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